

Home Assignment 1

1(a) Linearity

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, f linear. from linearity of f , denote $f(x) = Ax$

$g: \mathbb{R}^m \rightarrow \mathbb{R}^l$, g linear. similarly to f , $g(x) = Bx$

n.t.s $h(x) = g(f(x))$ is linear. ($h(x) = B(Ax) = BAx$)

Properties of linearity: Additivity: $f(x+y) = f(x) + f(y)$,
Homogeneity: $f(\alpha \cdot x) = \alpha \cdot f(x)$

Proof: Additivity: $h(x+y) = BA(x+y) = BAx + BAy = \underline{h(x) + h(y)}$

Homogeneity: $h(\alpha x) = BA(\alpha x) = \alpha BAx = \underline{\alpha \cdot h(x)}$

$\Rightarrow h$ is linear

1(b) Affinity

$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$, f affine. from affinity of f , denote $f(x) = Ax + a$

$g: \mathbb{R}^m \rightarrow \mathbb{R}^l$, g affine. similarly to f $g(x) = Bx + b$ } constants

n.t.s $h(x) = g(f(x))$ is affine. $h(x) = B(Ax + a) + b = BAx + Ba + b$

Proof: $h(x) = BAx + Ba + b$. Denote $BA = C$, $Ba + b = d$

$\Rightarrow h(x) = Cx + d \Rightarrow C$ is linear, and d is constant w.r.t x

$\Rightarrow \underline{h \text{ is affine}}$

2. Gradient Descent

a. obj function - $\min \left[f(x,y) = 10 \cdot (x-1)^2 + \frac{(y+1)^2}{10} \right]$.

$$\frac{\partial f}{\partial x} = 10 \cdot 2(x-1) = 20(x-1), \quad \frac{\partial f}{\partial y} = \frac{1}{10} \cdot 2 \cdot (y+1) = \frac{1}{5}(y+1)$$

$$\eta = LR \Rightarrow x_{k+1} = x_k - \eta \cdot \frac{\partial f}{\partial x} \Rightarrow x_{k+1} = x_k - \eta \cdot 20(x_k - 1) = \underbrace{x_k(1-20\eta)}_{r_x} + \eta \cdot 20$$

$$y := y - \eta \cdot \frac{\partial f}{\partial y} \Rightarrow y_{k+1} = y_k - \eta \cdot \frac{1}{5}(y_k + 1) = \underbrace{y_k(1 - \eta \cdot \frac{1}{5})}_{r_y} - \eta \cdot \frac{1}{5}$$

b. maximum learning rate

first let's compute the hessian:

$$\frac{\partial^2 f}{\partial x^2} = 20, \quad \frac{\partial^2 f}{\partial y^2} = \frac{1}{5}, \quad \frac{\partial^2 f}{\partial x \partial y} = 0 \Rightarrow H = \begin{pmatrix} 20 & 0 \\ 0 & \frac{1}{5} \end{pmatrix}$$

eigenvalues of H are $\lambda_1 = 20, \lambda_2 = \frac{1}{5} \Rightarrow \lambda_{\max} = 20$

$$\eta_{\max} = \frac{2}{\lambda_{\max}} = \frac{2}{20} = \underline{\underline{\frac{1}{10}}}. \quad \underline{\text{for any } \eta > \frac{1}{10} \text{ gradient descent will not converge}}$$

Convergence guaranteed iff $|r_x|, |r_y| < 1 \Rightarrow |r_x| < 1 \Rightarrow |1-20\eta| < 1$

$$\Rightarrow -1 < 1-20\eta < 1 \Rightarrow -2 < -20\eta < 0 \Rightarrow 0 < \eta < \frac{1}{10} \Rightarrow \underline{\underline{\max LR = \frac{1}{10}}}$$

c. Because took $\lambda_{\max} = 20$ which came from $\frac{\partial^2 f}{\partial x^2}$, so the GD in the direction of x should be more cautious. so the '10' in the objective function together with the '2' in the power, determine the max LR.

d. y takes the longest. that's determined by the smallest 2nd derivative. the convergence depends on $r_y = 1 - \eta \cdot \frac{1}{5}$ which is closer to 1 than $r_x = 1 - 20\eta$. the closer r is to 1 the slower the convergence.

3.

- $\text{pred_deg} := \text{pred_deg} \% 360$
- $\text{true_deg} := \text{true_deg} \% 360$
- $\text{diff} := |\text{true_deg} - \text{pred_deg}|$
- $\text{loss} := \min(\text{diff}, 360 - \text{diff})$
- return loss

4. Chain Rule

a) $\frac{\partial}{\partial x} f(x+y, 2x, z)$. Denote $u(x,y)=x+y$, $v(x)=2x$, $w(z)=z$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial x} + \frac{\partial f}{\partial w} \cdot \frac{\partial w}{\partial x} \quad \frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial x} = 2, \quad \frac{\partial w}{\partial x} = 0$$

$$\downarrow \quad \boxed{\frac{\partial f}{\partial u} + 2 \cdot \frac{\partial f}{\partial v}}$$

b) $F(x) = f_1(f_2(\dots f_n(x)\dots))$. $\frac{\partial F}{\partial x} = \frac{\partial f_1}{\partial f_2} \cdot \frac{\partial f_2}{\partial f_3} \cdot \frac{\partial f_3}{\partial f_n} \dots \frac{\partial f_n}{\partial x}$

$$\frac{\partial F}{\partial x} = \boxed{f_1'(f_2(\dots)) \cdot f_2'(f_3(\dots)) \dots f_n'(x)}$$

c) $F(x) = f_1(x, f_2(x, \dots f_{n-1}(x, f_n(x) \dots))$

$$\frac{\partial f_k}{\partial x} = \frac{\partial f_k}{\partial x} + \frac{\partial f_k}{\partial f_{k+1}} \cdot \frac{\partial f_{k+1}}{\partial x} \Rightarrow \boxed{\frac{\partial F}{\partial x} = \frac{\partial f_1}{\partial x} + \frac{\partial f_1}{\partial f_2} \left(\frac{\partial f_2}{\partial x} + \frac{\partial f_2}{\partial f_3} \left(\dots \frac{\partial f_{n-1}}{\partial x} + \frac{\partial f_{n-1}}{\partial f_n} \cdot f_n'(x) \dots \right) \right)}$$

d) $F(x) = f(x+g(x+h(x)))$ $u(x)=x+g(x+h(x))$, $v(x)=x+h(x)$

$$F(x) = f(u(x)) \Rightarrow \frac{\partial F}{\partial x} = \frac{\partial f}{\partial u} \cdot \frac{\partial u}{\partial x} = \frac{\partial f}{\partial u} \left(1 + \frac{\partial g}{\partial x} \right), \quad \frac{\partial g}{\partial x} = \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial g}{\partial v} \left(1 + \frac{\partial h}{\partial x} \right)$$

$$\boxed{\frac{\partial F}{\partial x} = \frac{\partial f}{\partial u} \left(1 + \frac{\partial g}{\partial v} \left(1 + \frac{\partial h}{\partial x} \right) \right)}$$